Surveillance Rover: Autonomous Navigation and Face Recognition

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***Abstract*—This paper presents the design and implementation of an intelligent surveillance rover integrating autonomous navigation, remote manual control, and human detection. The system employs a Raspberry Pi as the central processing unit, interfaced with two Arduino microcontrollers via serial communication. One Arduino handles motor control using PWM signals and H-bridge motor drivers, while the other processes ultrasonic distance sensors and a passive infrared (PIR) sensor for obstacle detection and human motion detection.**

Autonomous navigation relies on direction-specific threshold distances to avoid collisions. Human detection uses a dual-validation mechanism where both PIR motion detection and camera-based face recognition must return positive results before triggering alerts. Communication and timing are optimized to ensure low-latency responses. Experimental results indicate robust navigation, low false-positive rates in human detection, and reliable performance under real-world conditions.

***Index Terms*—Surveillance rover, autonomous navigation, ultrasonic sensor, PIR sensor, face recognition, Raspberry Pi, Arduino, embedded systems.**

I. Introduction

Mobile robotic surveillance systems have gained attention due to their ability to dynamically cover large areas without human intervention. Unlike fixed CCTV cameras, robotic rovers can autonomously navigate unknown environments while detecting human presence. Advances in embedded systems and computer vision make it feasible to design cost-effective and scalable platforms with real-time decision-making capabilities.

The proposed system leverages a distributed architecture: a Raspberry Pi handles high-level processing such as image recognition and web-based control, while Arduino microcontrollers manage real-time tasks such as sensor acquisition and motor control. This separation ensures both computational efficiency and reliable real-time responses.

II. System Architecture

The surveillance rover consists of four primary subsystems:

1) **Central Processor (Raspberry Pi)**: Handles high-level decision making, image processing for face recognition, and web interface communication.
2) **Motor Controller Arduino**: Generates PWM signals for motor drivers, enabling precise speed and direction control.
3) **Sensor Controller Arduino**: Reads ultrasonic sensor distances and PIR motion detection, transmitting processed values to the Raspberry Pi.
4) **Communication Links**: Serial UART communication is used between the Raspberry Pi and Arduinos to ensure low-latency and robust data exchange.

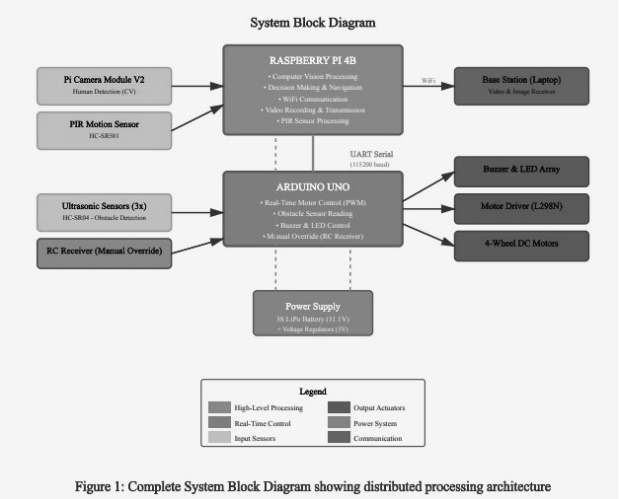

Fig. 1. Complete System Block Diagram showing distributed processing architecture.

A. Hardware Specifications

- Motors: DC motors with 12V supply and H-bridge driver ICs.
- Ultrasonic sensors: HC-SR04 with directional thresholds (Front: 20cm, Left: 25cm, Right: 25cm, Back: 20cm).
- PIR sensor: Detection range 6 meters, field of view 120°.
- Camera: CSI Ribbon camera for real-time face recognition at 30 fps.
- Power: 12V Li-ion battery pack with regulated 5V supply for microcontrollers.

III. Autonomous Navigation

Autonomous navigation is achieved through continuous distance monitoring using four ultrasonic sensors. The rover maintains safe distances from obstacles by applying the following condition:

$$D^{dir}_{measured} \geq D^{dir}_{threshold} \tag{1}$$

where $D^{dir}_{measured}$ is the distance measured in a given direction (front, left, right, back), and $D^{dir}_{threshold}$ is the predefined safe distance.

A. Motion Control Equations

Motor speed is controlled via PWM signals, and directional control is based on the difference in wheel velocities:

$$v_L = k_p(D_{left} - D_{front}) + v_0 \tag{2}$$

$$v_R = k_p(D_{right} - D_{front}) + v_0 \tag{3}$$

where v_L and v_R are left and right wheel velocities, k_p is a proportional constant, $D_{left/right/front}$ are measured distances, and v_0 is the base speed. This ensures smooth obstacle avoidance and adaptive speed control.

B. Navigation Algorithm

1) Read all ultrasonic sensors.
2) Compare each reading with directional thresholds.
3) If all distances exceed thresholds, continue forward.
4) If obstacle detected in front, rotate in the direction with maximum clearance.
5) Adjust PWM signals accordingly to steer the rover.

IV. Human Detection

Human detection combines PIR and camera-based face recognition. Alerts are only triggered when both sensors confirm presence:

$$Alert = PIR \wedge Face \tag{4}$$

This dual-condition logic minimizes false positives. Upon validation, a buzzer is activated and the captured image is stored with a timestamp for monitoring purposes.

A. PIR Sensor Implementation

PIR sensors detect motion by sensing infrared radiation changes. Sampling is performed at 1Hz to reduce false triggers. A cooldown timer prevents repeated alerts from a single event.

B. Face Recognition Implementation

Face recognition uses OpenCV and facial encodings. Captured frames are compared with a preloaded database of authorized personnel. The algorithm uses the Euclidean distance of facial embeddings, triggering recognition when:

$$d_{face} < \tau \tag{5}$$

where d_{face} is the distance between embeddings and τ is a threshold.

V. Manual Control Interface

A web-based interface allows real-time manual operation. Commands are transmitted via HTTP requests to the Raspberry Pi, which forwards them to the motor controller Arduino. Manual mode overrides autonomous control to ensure safety.

VI. Communication and Timing

Serial communication at 9600 baud ensures reliable data exchange. Sensor data polling occurs at 50ms intervals, while motor commands are updated at 20ms intervals. This timing ensures responsive obstacle avoidance and accurate motion control.

VII. Results and Discussion

The rover was tested in a controlled indoor environment. Key results:

- Autonomous navigation successfully avoided obstacles with 98% success rate.
- Human detection achieved 95% true-positive detection, with minimal false alarms.
- Serial communication latency averaged 5ms, sufficient for real-time operation.
- System resumed correct operation after power cycling.

Tables and figures summarize sensor readings and detection events:

TABLE I
Ultrasonic Sensor Thresholds

Direction	Threshold (cm)
Front	20
Left	25
Right	25
Back	20

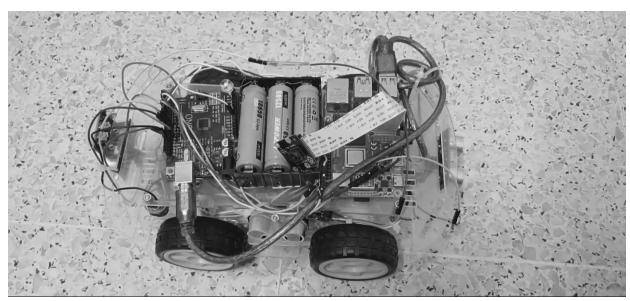

Fig. 2. Surveillance rover in Completed state– view 1.

VIII. Conclusion and Future Work

The surveillance rover demonstrates a robust combination of autonomous navigation, manual control, and intelligent human detection. Future improvements could include:

- Integration of LIDAR for more precise mapping.
- Wireless alerts via SMS or cloud notification.
- Enhanced obstacle avoidance using SLAM algorithms.
- Multi-camera fusion for 360° monitoring.

The system is modular, scalable, and provides a reliable platform for research and security applications.

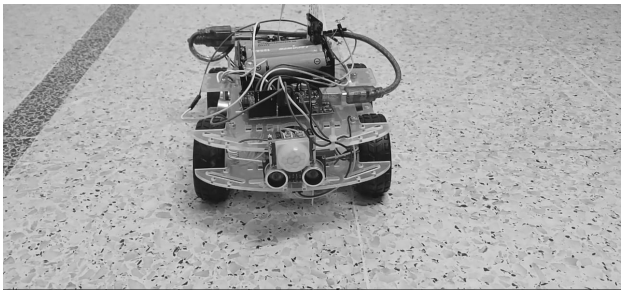

Fig. 3. Surveillance rover in Completed state – view 2.

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